

Coexistence From First Principles

A generative kernel for the coexistence template

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The thesis: The six primitives of [The Coexistence Template](#) — the medium O , units U , relationships R , values V , composition κ , and transmission τ — can be generated from a kernel of three: *satta* as the omnipresent ground, *activity* carrying the inseparable triad of effort, motion, and result, and *saturation* as the single primitive relation between them. Units are bounded activity whose motion-path closes; relationships are definite couplings of triads; value is what a coupling realises; composition is a coupling that closes a new boundary; transmission is the composition method recurring as constitutive input. The construction succeeds exactly where the texts assert generatively and stops exactly where they supply content — and the paper’s central result is a **recovery audit** classifying every clause of the template as *derived* from the kernel, *definitional*, or *substantive textual data*, with the irreducible remainder stated up front rather than discovered later.

Madhyasth Darshan’s exposition is layered: *Samadhanatmak Bhautikvad* opens with fully general assertions about units, energy, and activity before any particular order is discussed, and *Manav Vyavahar Darshan* derives its human programme from the same general ground. [The Coexistence Template](#) articulated that ground as a tier-neutral structure with six primitives, and its generative-order overview (§3) already reads the clauses as one chain: energy, activity, unit, mutuality, value, composition, transmission. This paper takes the next step the overview invites: state the head of the chain as a formal kernel and *construct* the rest, so that what the template postulates becomes, where the texts warrant it, a theorem — and where they do not, a clearly labelled datum. The discipline follows the conservativity standard adopted in [Category Theory Explained](#) §6.16.4, and the recovery audit here is the first execution of the independence-audit programme both papers announce.

1. Why a kernel, and what counts as success

A formalisation earns reduction only if it can say precisely what was gained and what was assumed. The template’s six primitives are textually faithful, but they conceal dependency structure: units are *defined* as bounded activity, relationships are *defined* over units, value is *defined* as what flows in relationships — the definitions lean on one another in a definite order the primitive list flattens. The kernel makes the order load-bearing. Success is measured against the template itself, clause by clause: a construction counts as a **derivation** when the template clause follows from kernel axioms plus definitions; as **definitional** when it merely

names structure the kernel already contains; and as **substantive data** when the texts assert content no kernel manipulation could produce — the specific six virtues, the four carriers of transmission, the restriction of evaluation to *jeevan*. The audit (§5) assigns one of these three statuses to every clause D1–D13 and L1–L7, and §7 states the irreducible remainder plainly.

Two cautions govern everything that follows. The texts *assert* piecewise; this paper *derives* — every derivation is this paper’s construction, warranted by citations but not identical with them, and the template remains the textual control against which the kernel is judged. And a reduction can flatten: the review of Meena’s topos-theoretic formalisation ([Category Theory Explained](#) §6.16.6) found a single-sort predicate ledger discarding precisely the structure a formalisation of this darshan must keep. The kernel avoids that fate only by keeping activity — not featureless points — as its dynamical primitive, and by refusing to derive what the texts present as content.

2. The kernel

2.1 Three primitives

O (*satta*, Omnipresence). The indivisible, omnipresent, actionless reality: “Space is uniform in all dimensions... this reality is omnipresent” (JV p. 31). O is state-complete while everything else is state-dynamic (MVD p. 12); it is energy in equilibrium and knowledge-natured (*gyan*), the intelligibility-ground of whatever is saturated in it (MVD p. 11).

A (activity). The dynamical primitive, carrying its triad internally: every activity has **effort** (*shram*), **motion** (*gati*), and **result** (*parinam*) as aspects, and each aspect is the joint form of the other two — “The outcome of motion and result is effort. The outcome of effort and motion is the result. The outcome of result and effort is motion” (MVD p. 105). The combined activity is *mool cheshta*, basic impulsion (MVD p. 40). Nothing about units, boundaries, or relationships is presupposed here: A is activity as such.

Saturation. The single primitive relation: every activity is in O — soaked, surrounded, submerged — and O is not an activity. “There is no provision in existence to separate units of nature from Omnipresence” (JV p. 18). Saturation is constitutive co-location, not containment of one thing inside another thing of the same kind.

2.2 Kernel axioms

K1 (Energisation). Every activity is energy-replete through saturation, and activity is neither created nor destroyed. The texts state the equation directly:

“Nature is the set of countless units. The fundamental unit of nature is the atom. Every unit in its atomic state is active as orderliness, because it has inherent energy due to being saturated in Omnipotence. There is no unit or particle in nature which is not replete with energy. Every unit is active because of this itself. This activeness is seen in every unit in the form of effort, motion and result.”

— SB p. 69

“Unit + Energy fullness = Activeness” (SB p. 69): one uniform energy manifests as the triad. Conservation — the template’s A3 — becomes a corollary rather than an axiom: results transform activity, they never annihilate it (JV p. 127).

K2 (Boundedness). Activity clusters into bounded loci, and the bound is not imposed from outside — it is the closure of the activity’s own motion-path:

“Every atom’s activity is along with the motion-path of its particles. This motion-path itself is that atom’s line of regulation and also its boundary.”

— SB p. 58

The same closure that bounds also regulates: boundary and discipline are one structure, which is why the template’s regulation ladder can begin inside the unit rather than descending on it.

K3 (Coupling). Activities couple: one activity’s results enter another’s as input, and the texts name the coupling’s contribution explicitly —

“The development and order of matter depend on that unit’s motion, effort, and pressure of its mutuality - by which integration, disintegration, and result take place.”

— MVD p. 114

The definite form of coupling is complementarity of deficiency and surplus — development progression operates “in the atom in the form of hungry and overfull atoms” (MVD p. 8). Coupling is primitive in form; *which* couplings are admissible is content the texts supply case by case.

K4 (Recognition-as-law). Saturation provisions recognition: “Regulation itself becomes clear in the form of law. Consequently, there is provision in every unit for recognising one another based on law” (SB p. 57). Recognition is therefore not an added cognitive faculty at the bottom tiers; it is the lawfulness of coupling itself.

K5 (Completeness orientation). Each triad aspect has a completeness goal:

“Every atom continues to undergo insertion and expulsion of particles until constitutional completeness, because every atom exists in the form of activity, and every activity has an inseparable presence of effort, motion, and result. The result is meant to attain the goal of immortality, the effort is meant to attain the goal of restfulness, motion is meant to attain the goal of destination. This goal itself becomes evident in the form of development progression and development.”

— SB p. 71

Development is thereby directed without being an optimisation: the goals are terminal states (immortality, restfulness, destination), not maxima — the distinction [The Coexistence Template](#) §5 defends at length.

3. The constructions

3.1 Units from bounded activity

Define a **unit** as a maximal bounded cluster of activity whose motion-path closes (K2). Everything the template’s D1 asserts follows in form: a unit is “a bounded cluster of activity”; insentient units are active within their bounds, sentient units beyond them (MVD p. 33) — in kernel terms, the closure can bound the activity’s *extent* without bounding its *reach*. The signature $\text{sig}(u) = \langle \text{roop}, \text{gun}, \text{svabhav}, \text{dharma} \rangle$ splits by status: form (*roop*) is the geometry of the closed motion-path — derived; properties, essential nature, and *dharma* are what the texts assert about each order’s activity — data organised by the construction, not produced by it. This is the kernel’s strongest result: countability, boundedness, and the unit/medium distinction (O has no motion-path, hence is no unit) all fall out of K2 and saturation.

3.2 Relationships from coupling

Define a **relationship** as a definite coupling of triads (K3) whose profile is fixed by the coupled units’ orders and signatures; an **association** is a coupling whose profile is adopted rather than fixed — the template’s D3 distinction recovered as the modality of the coupling. The expectation profile $E(r)$ is the complementarity of deficiency and surplus between the coupled activities. Recognition ρ is the lawfulness of the coupling (K4); fulfilment ϕ is the coupling executing — results actually delivered into the partner’s activity. The template’s L1 (universal recognition-fulfilment) is then derived for the definite orders, and L2’s “no struggle” reading is the observation that coupling’s definite form is complementary, with the textual claim that *no* inter-unit dynamics is conflictual (JV p. 157) retained as assertion rather than theorem.

3.3 Value from participation

Define the **value** of a unit in a relationship as what its coupling realises: the contribution its triad makes to the partner's activity — essentiality as “participation-as-value in mutuality” (SB p. 50). The construction produces the *slot*: value as the content of coupling, real at every tier because coupling is. What fills the slot is data: the six kinds of value, the nine established values and their generative order, the distinction of *jeevan* values from human values ([The Coexistence Template](#) D4). A kernel that claimed to derive kindness from motion-paths would be exactly the over-derivation this paper exists to avoid.

3.4 Composition from closure of coupling

Define **compound composition** as a coupling that closes a *new* joint motion-path: the coupled activities cease to maintain separate closures and present one bounded activity with a new signature. Define **mixture** as coupling without new closure — the components' own boundaries persist. This recovers the template's D8 two-mode distinction (MVD p. 42) from $K_2 + K_3$ alone, and L6 (iteration) becomes immediate: the output of compound composition satisfies the definition of a unit, so every construction applies to it in turn. That the *sentient threshold* is reached along this path — constitutional completeness as a specific compound configuration (SB pp. 55, 59) — is data about which closure is special, not a consequence of closure in general.

3.5 Transmission from method persistence

Define **transmission** as the recurrence of a composition method: the assembly's *rachna vidhi* — itself a result of the assembly's activity — entering as constitutive input in successive member-units, “the method of formation is inherent in the pran sutra situated in the cell” (JV p. 82). The construction gives τ 's *form*: a result whose content is a method, recurring across member turnover. It does not give τ 's carriers — constitution, seed, lineage, education-*sanskar* (JV p. 48; MVD p. 93) — which ascend with the orders as textual data, nor the knowledge-order asymmetry that only understanding flows in tradition (JV p. 49). C5 is a constructor schema, the weakest of the five constructions, and is flagged as such in the audit.

4. Orders, jeevan, and the reflexive triad

The four orders classify how a unit's triad maintains its definiteness: by result-conformance, seed-conformance, species-conformance, or *sanskar*-conformance (JV p. 48; SB p. 236). In kernel terms an order is a **persistence regime** — the way a bounded activity's method survives — which is why the order taxonomy and the transmission carriers align row by row. The four-fold list itself, and each order's characteristic *svabhav* and *dharma*, are data.

Jeevan is defined in the kernel as a unit whose result-aspect has reached its K5 goal: immortality of result — *gathanpurnata*, after which “both effort and motion become inexhaustible” (SB p. 61; the strengths and powers of the constitutionally complete atom, SB p. 80). The

knowledge order's additional capabilities are then one move: **the triad applied reflexively**. *Chitta*'s eight activities of visualisation include effort, motion, and result themselves (MVD p. 327) — a *jeevan* is an activity that grasps activities. Knowing and believing, evaluation and choice (JV p. 70) are second-order operations on couplings: evaluation μ assesses the value a coupling realises against the value inherent in it, and justice is the closed cycle recognise $\rightarrow$ fulfil $\rightarrow$ evaluate $\rightarrow$ mutual satisfaction (MVD p. 311). The template's knowledge-order discontinuity (P3) acquires a kernel formulation: **reflexive triads must achieve the definiteness that non-reflexive triads have by constitution** — a second-order coupling can misfire (over-, under-, mis-evaluation, MVD p. 38) precisely because it is mediated by grasping, and *kshobh* at this tier is the frustration of effort that “itself is their yearning for restfulness” (MVD p. 104), driving the awakening progression the way deficiency drives bonding below.

The reflexivity reading is this paper's interpretive contribution, not a quotation. What the texts assert is the list of additional activities and their location in *jeevan*; that the list is unified as the triad turned on triads is a construction — flagged accordingly in the audit, and offered because it explains with one device why evaluation exists only where constitutional completeness does.

5. The recovery audit

The kernel is adequate only if the template survives translation. The audit below assigns each clause of [The Coexistence Template](#) §3 one of three statuses: **derived** (follows from K1–K5 plus the constructions), **definitional** (a conservative name for kernel structure), or **data** (textual content the kernel organises but cannot produce).

Template clause	Status	Notes
A1 (saturation)	kernel primitive	The saturation relation itself
A2 (energisation and regulation)	kernel axiom	K1 + K4
A3 (conservation)	derived	Corollary of K1: results transform, never annihilate
A4 (no isolation)	derived, with one assumption	Totality of coupling on units; the texts assert it (JV p. 43), the kernel needs it as a clause of K3
A5 (inherent orderliness)	derived	K2's closure is the unit's own orderliness; participation from K3
D1 (unit)	derived	C1; <i>roop</i> from boundary geometry
D1a (order signature)	data	Per-order <i>svabhav</i> and <i>dharma</i>
D1b (activity triad)	kernel primitive	The structure of A

Template clause	Status	Notes
D2 (orders)	definitional + data	Persistence regimes as the classifying notion; the four-fold list is data
D2a (regulation ladder)	definitional	Reading order of K1, K4, D2 — as the template itself says
D2b (inward regulation)	data	Faculty structure of <i>jeevan</i> ; the mediative pattern parallels K2's nucleus but is not derived by it
D3 (relationship and association)	derived	C2; the definite/adopted split as modality of coupling
D4 (value)	derived slot + data	C3; the six kinds and nine established values are data
D5 (recognition and fulfilment)	derived	K4 + coupling execution
D5a (fulfilment capacity)	data	The triad (ksh, yog, pat) is textual; its gating role is derived from C2
D6 (evaluation)	data	The restriction to <i>jeevan</i> is irreducible; the reflexivity reading (§4) organises but does not derive it
D7 (justice)	definitional	The name of the closed second-order cycle
D7a (trust)	data	Value-level name of successful φ
D8 (composition, two modes)	derived	C4
D9 (natural and excited state)	derived	K5: closure toward goals versus charged non-closure
D10 (transmission)	schema + data	C5's form derived; carriers data
D11 (planes and transitions)	derived orientation + data	K5 gives the direction; plane names and T1 irreversibility are data
D12 (knowledge registers)	data	Organised by the reflexivity reading
D13 (evidence chain)	data	Knowledge-order specific; no kernel derivation claimed

Template clause	Status	Notes
L1 (recognition-fulfilment)	derived	For definite orders; achieved mode at the knowledge order per §4
L2 (complementarity, not struggle)	derived form + assertion	Complementary form from K3; universality of non-conflict is textual
L3 (assembly by complementary need)	derived form + data	The guard's content — which complementarities are admissible — remains data, as the template's §7 and Category Theory Explained §6.9.1 already concluded
L4 (persistence $\Leftrightarrow$ fulfilment)	derived	K5 + C2: natural state as coupling closing
L5 (transmission by order)	data	The carrier ladder
L6 (closure/iteration)	derived	C4's output is a unit
L7 (self-evidencing)	data	With the reflexive formulation of §4 as its organising frame

Independence of the kernel axioms is witnessed by countermodels in the intended informal semantics: without K2 there is undifferentiated activity and no unit — nothing for the rest of the template to be about; without K3 units are windowless and value has no slot; without K4 coupling is arbitrary and the definite orders lose their lawfulness; without K5 development has no direction and the natural/excited distinction collapses; without K1 activity floats free of the medium and the template's first ontological layer (saturation before *sambandh*) is lost. Formalising these countermodels — and mechanising the whole audit — is the continuation this paper's discipline makes possible but does not yet execute.

The headline: of the template's thirty audited clauses, roughly a third are derived, a handful are definitional, and the remainder is data — which is not a defect but the honest shape of a philosophy whose ground is asserted from realisation and whose content is taught. The kernel's value is that the boundary now runs through named clauses instead of through an undifferentiated primitive list.

6. Mathematical form: boxes, wiring, and collapse

The kernel suggests its own mathematics, different from the colimit vocabulary of [Category Theory Explained](#) §6.8–6.10 and closer to the operadic treatment of open systems. A unit is a **box with typed ports**: its deficiencies and surpluses, read off its triad. A relationship is a **wiring** between ports — results flowing out of one box into another, the pressure of mutual-ity made diagrammatic. Compound composition is the **collapse** of a wired diagram into a single box with a new interface: the operation operads axiomatise, and exactly the template's

L6 — the collapsed box is again a box, so wiring and collapse iterate up the tiers. Mixture is wiring without collapse. The L3 guard becomes a *typing constraint on wirings* — only complementary port-pairs connect — rather than an external predicate on spans, which is a structural improvement on the `AdmissibleComp` device even though the port types themselves remain textual data. Transmission is the least natural fit, as it was for colimits: a collapsed box’s wiring pattern re-instantiated in fresh boxes is algebra-like structure over the operad, and its correct formulation is left open. Petri-net bookkeeping ([Category Theory Explained](#) §6.10) remains the right tool for the resource dynamics *inside* this picture; the operadic layer organises what the boxes and wirings *are*. Developing this sketch into a theorem-bearing formalism is the natural sequel.

7. What the kernel does not capture

The irreducible remainder, stated as commitments rather than residue. **O and saturation** are primitives here as everywhere in this collection: the kernel begins from them and cannot ground them. **The triad structure of A** is taken from the texts whole — the kernel does not explain why activity has exactly these three aspects. **The content of admissibility** (which complementarities bond), **the transmission carriers**, **the six kinds of value**, and **the four orders’ signatures** are taught content. **Evaluation’s restriction to jeevan** is asserted, and the reflexivity reading organises it without deriving it. **Jeevan’s substantiality** — the structuralism-versus-substance tension of [Category Theory Explained](#) §6.13.1 — is untouched: a kernel of activities re-describes *jeevan* functionally exactly as arrows and predicates did. And the **samadhi warrant** for the axioms (MVD p. 7) remains meta to any formalism, this one included.

Open problems, collected. The formal countermodels for K1–K5 and mechanisation of the recovery audit. The operadic formulation of §6, especially transmission-as-algebra. Whether the reflexivity reading of the knowledge order can be sharpened into a definition with consequences — for instance, deriving the possibility of the three evaluation failures from second-order coupling structure. And whether the kernel’s derivation of D9 from K5 can be connected formally to the stationarity comparison in [Restfulness and Least Action](#) §6.2.

8. Conclusion

Read generatively, Madhyasth Darshan’s ground is small: one omnipresent reality, activity bearing effort, motion, and result, and the saturation of the second in the first. From that kernel, the unit, the relationship, the value-slot, both modes of composition, and the form of transmission are constructions — and the tiered world of the template, from atoms to the possibility of undivided society, is the iteration of those constructions under the completeness orientation the texts assert for every activity. What the kernel cannot produce, it now names: the taught content of the darshan, clause by clause. The template articulated the structure; this paper shows how much of it the structure’s own head generates, and holds the

rest out in the open where the independence audit — and eventually a proof assistant — can reach it.

References

Madhyasth Darshan (primary sources)

- **MVD** — Nagraj, A. [*Madhyasth Darshan \(Manav Vyavahar Darshan\)*](#). English version 4.0. Cited: state-complete/state-dynamic and evidence chain (p. 12); *gyan* as intelligibility-ground (p. 11); basic impulsion (p. 40); hungry/overfull atoms (p. 8); joint form of effort, motion, and result (p. 105); mutuality pressure in integration, disintegration, and result (p. 114); justice as closure (p. 311); evaluation failures (p. 38); insentient/sentient activity bounds (p. 33); *chitta*'s eight visualisation activities (p. 327); frustration of effort as yearning for restfulness (p. 104); knowing and believing (via JV p. 70 parallel); samadhi-samyama warrant (p. 7). Page numbers refer to this edition.
- **SB** — Nagraj, A. [*Samadhanatmak Bhautikvad \(Resolution Centred Materialism\)*](#). English translation by Rakesh Gupta. Also at <https://www.youtube.com/playlist?list=PL69PCoz1OQWodhshZoXv3KtZ7ajJOIpgv> (bilingual Hindi and English). Cited: unit + energy fullness = activeness (p. 69); motion-path as regulation line and boundary (p. 58); effort–motion–result completeness goals (pp. 58, 71); regulation as law with recognition provision (p. 57); essentiality as value (p. 50); constitutional completeness and inexhaustibility (pp. 55, 59, 61, 80); order conformance regimes (p. 236). Page numbers refer to this edition.
- **JV** — Nagraj, A. [*Jeevan Vidya: An Introduction \(ju-parichaya\)*](#). English translation, version 3.0. Cited: Space uniform and omnipresent (p. 31); no provision to separate units from Omnipresence (p. 18); nothing isolated (p. 43); order conformance (p. 48); understanding flows in tradition (p. 49); *rachna vidhi* in the *pran sutra* (p. 82); no conflict in existence (p. 157); knowing and believing as additional activities (p. 70); existence neither increases nor decreases (p. 127). Page numbers refer to this edition.

Formal approaches to Madhyasth Darshan

- **MD-TOPOS** — Meena, B. [*Minimal Decidable Site for the Madhyasth–Darshan Classifying Topos via Single-Flag Morleyisation*](#). Zenodo preprint, 2025; DOI [10.5281/zenodo.16786431](https://doi.org/10.5281/zenodo.16786431). Cited: minimality and independence proofs for a predicate ledger — the methodological precedent for the recovery audit of §5, with the contrast drawn in §1.

Related studies in this collection

- [*The Coexistence Template*](#) — the textual articulation this paper's kernel must recover: primitives, D1–D13, L1–L7, and the generative-order overview (§3) whose chain the

kernel formalises

- [Category Theory Explained](#) — categorical notation, conservativity discipline and independence-audit programme (§6.16.4), Petri layer for the triad (§6.10), structuralism-versus-substance boundary (§6.13.1)
- [The Ontology of Coexistence](#) — ontological exposition of saturation, the regulation ladder, orders, planes, and progressions
- [Restfulness and Least Action](#) — the energetic registers of the triad and the stationarity comparison (§6.2) that K5's derivation of D9 connects to

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